THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 4: **Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student connects a bulb, a switch and a cell but the bulb does not glow even when the switch is closed. The bulb holder is loose. The fault is a: 1

(a) break in the circuit **(b)** magnetic effect
(c) overload **(d)** short circuit

2. A toaster, an iron and a heater use the same effect of current, which is: 1

(a) heating **(b)** magnetic **(c)** chemical **(d)** light

3. The magnetic effect of current is shown by a: 1

(a) bulb glow **(b)** wire melting
(c) bubble formation **(d)** compass needle deflection

4. Two wires of equal length and thickness, one nichrome and the other copper, carry the same current. Which gets hotter? 1

(a) copper **(b)** nichrome **(c)** both equally **(d)** neither

5. A scrap yard crane drops the iron when: 1

(a) the current is switched off **(b)** the current is on
(c) the iron is heavy **(d)** the wire is long

6. A student tests materials in a circuit: bulb glows with a coin, key and iron nail; not with eraser, wood and glass. The materials that conduct are: 1

(a) only the iron nail **(b)** all of them
(c) eraser, wood and glass **(d)** coin, key and iron nail

7. A compass needle placed near a battery wire does not deflect. A likely reason is: 1

(a) the wire is copper **(b)** the cell is new
(c) the needle is a magnet **(d)** the switch is open

8. A student connects two cells in series in a circuit with a bulb. The bulb: 1

(a) goes out **(b)** does not glow **(c)** glows brighter **(d)** glows dimmer

9. A wire carrying current produces a field in the form of: 1

(a) straight lines along the wire **(b)** circles around the wire
(c) no pattern **(d)** dots

10. An electromagnet lifts 10 clips with 20 turns. With 40 turns of the same wire and cell, it will lift: 1

(a) fewer clips **(b)** no clips
(c) the same number of clips **(d)** more clips

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A bulb does not glow in a circuit. List two possible faults and how you would check them. 2

12. Why does an electromagnet pick up more pins when the number of turns is increased? 2

13. A thin and thick nichrome wire of equal length have the same current. Which heats more and why? 2

14. Why is copper used for connecting wires and nichrome for heating coils? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student made an electromagnet but it can't pick up pins. List three possible reasons and remedies. 3

16. Explain how an MCB works. What is the advantage over a fuse? 3

17. Why does an electric kettle have a thick copper cord but a thin nichrome heating coil? 3

OR

Explain why long thin wires are not used for the household wiring but short thick wires with insulation are. 3

18. Why are electric appliances fitted with a three-pin plug having an earth pin? State the role of earth. 3

OR

List three reasons why electrical accidents occur in homes. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe an activity to show that a current-carrying wire behaves like a magnet. Mention the observations when the current is reversed and when the compass is placed in different positions. 5

OR

Describe the safety systems in a home electrical circuit: fuse, MCB, earthing, insulation, switches, and say what each protects against. 5

20. Explain the construction and working of an electric bell with a flow chart in words. 5

OR

Explain how an electromagnet is used in a crane to lift scrap iron and to drop it. Why is an electromagnet suitable here? 5

END OF QUESTION PAPER — Total: 40 Marks